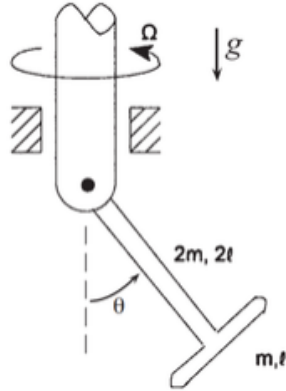


Problem 1

Problem 1



A bar of mass m , length ℓ is welded to a bar of mass $2m$, length 2ℓ to form the T shown. The body is pinned to a post that rotates at the constant rate Ω .

The moment of inertia tensor for the T-shaped bar is given by,

$$I = \begin{bmatrix} \frac{1}{12}m\ell^2 & 0 & 0 \\ 0 & \frac{20}{3}m\ell^2 & 0 \\ 0 & 0 & \frac{27}{4}m\ell^2 \end{bmatrix}$$

in the body frame $\{\hat{b}\}$ along the principal axes of the body, centered at the pivot point, with $\hat{b}_1$ pointing along the length- 2ℓ bar (down and to the right), $\hat{b}_2$ pointing up and to the right parallel to the length- ℓ bar, and $\hat{b}_3$ pointing out of the page.

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- (a) Calculate the angular velocity ω and angular momentum H of the pendulum, each expressed in the body frame.

Let $\hat{n}_2$ be the vertical spin axis of the post, fixed in the inertial frame, so $\vec{\Omega} = \Omega \hat{n}_2$.

$$\hat{n}_2 = -\cos \theta \hat{b}_1 + \sin \theta \hat{b}_2$$

$$\vec{\omega} = \Omega \hat{n}_2 + \dot{\theta} \hat{b}_3 = -\Omega \cos \theta \hat{b}_1 + \Omega \sin \theta \hat{b}_2 + \dot{\theta} \hat{b}_3$$

$$\vec{H} = I\vec{\omega} = -\frac{1}{12}m\ell^2\Omega \cos \theta \hat{b}_1 + \frac{20}{3}m\ell^2\Omega \sin \theta \hat{b}_2 + \frac{27}{4}m\ell^2\dot{\theta} \hat{b}_3$$

$${}^B \frac{d\vec{H}}{dt} = \frac{1}{12} m \ell^2 \Omega \dot{\theta} \sin \theta \hat{b}_1 + \frac{20}{3} m \ell^2 \Omega \dot{\theta} \cos \theta \hat{b}_2 + \frac{27}{4} m \ell^2 \ddot{\theta} \hat{b}_3$$

- (b) Calculate the rate of change of angular momentum, ${}^N \frac{dH}{dt}$.

$${}^N \frac{d\vec{H}}{dt} = {}^B \frac{d\vec{H}}{dt} + \vec{\omega} \times \vec{H}$$

$${}^N \frac{d\vec{H}}{dt} = \frac{1}{6} m \ell^2 \Omega \dot{\theta} \sin \theta \hat{b}_1 + \frac{40}{3} m \ell^2 \Omega \dot{\theta} \cos \theta \hat{b}_2 + \left(\frac{27}{4} m \ell^2 \ddot{\theta} - \frac{79}{12} m \ell^2 \Omega^2 \sin \theta \cos \theta \right) \hat{b}_3$$

- (c) Compute the applied moment due to gravity, treating that moment as two forces applied at the center of mass of each bar. Note that there is also a torque applied about the vertical axis to keep the shaft spinning at a constant rate Ω .

$$\vec{r}_{2m} = \ell \hat{b}_1, \quad \vec{r}_m = 2\ell \hat{b}_1, \quad \hat{g} = \cos \theta \hat{b}_1 - \sin \theta \hat{b}_2:$$

$$\vec{F}_{2m} = 2mg \cos \theta \hat{b}_1 - 2mg \sin \theta \hat{b}_2, \quad \vec{F}_m = mg \cos \theta \hat{b}_1 - mg \sin \theta \hat{b}_2$$

$$\sum \vec{M} = \ell \hat{b}_1 \times \vec{F}_{2m} + 2\ell \hat{b}_1 \times \vec{F}_m = -2\ell mg \sin \theta \hat{b}_3 - 2\ell mg \sin \theta \hat{b}_3$$

$$\sum \vec{M} = -4mg\ell \sin \theta \hat{b}_3$$

(The $\hat{b}_1, \hat{b}_2$ components of part (b) are balanced by the pin/motor reaction.) Only the $\hat{b}_3$ component is used in the equation of motion because the pin axis is $\hat{b}_3$, and the motor torque needed to hold Ω constant acts about the vertical axis, which lies entirely in the $\hat{b}_1$ - $\hat{b}_2$ plane, so $\hat{b}_3$ is the only component of the moment equation free of an unknown reaction/motor torque.

- (d) Find the equation of motion for θ , using ${}^N \frac{dH}{dt} = \Sigma M$. At what critical value of Ω does a bifurcation occur?

$$-4mg\ell \sin \theta = \frac{27}{4} m \ell^2 \ddot{\theta} - \frac{79}{12} m \ell^2 \Omega^2 \sin \theta \cos \theta$$

$$\ddot{\theta} = \frac{79}{81} \Omega^2 \sin \theta \cos \theta - \frac{16}{27} \frac{g}{\ell} \sin \theta$$

Small θ :

$$\ddot{\theta} \approx \left(\frac{79}{81} \Omega^2 - \frac{16}{27} \frac{g}{\ell} \right) \theta$$

Bifurcation when the coefficient vanishes:

$$\frac{79}{81} \Omega^2 = \frac{16}{27} \frac{g}{\ell} \implies \Omega_{crit} = \sqrt{\frac{48}{79} \frac{g}{\ell}}$$

This is a pitchfork bifurcation: for $\Omega < \Omega_{crit}$, $\theta = 0$ is the only equilibrium and it's stable, while for $\Omega > \Omega_{crit}$ it loses stability and two new symmetric equilibria appear at $\theta = \pm \arccos(\Omega_{crit}^2 / \Omega^2)$ (the full nonlinear $\ddot{\theta} = 0$ roots), consistent with the sign-flip of the linearized coefficient above.

(e) Plot the phase portrait before and after Ω_{crit} .

$$\theta_1 = \theta, \theta_2 = \dot{\theta}:$$

$$\begin{cases} \dot{\theta}_1 = \theta_2 \\ \dot{\theta}_2 = \frac{79}{81}\Omega^2 \sin \theta_1 \cos \theta_1 - \frac{16}{27}\frac{g}{\ell} \sin \theta_1 \end{cases}$$

$$J = \begin{bmatrix} 0 & 1 \\ \frac{79}{81}\Omega^2 \cos 2\theta_1 - \frac{16}{27}\frac{g}{\ell} \cos \theta_1 & 0 \end{bmatrix}$$

At $\theta_1 = 0$: $J_{21} = \frac{79}{81}\Omega^2 - \frac{16}{27}\frac{g}{\ell}$, giving imaginary eigenvalues (center) for $\Omega < \Omega_{crit}$, real opposite-sign (saddle) for $\Omega > \Omega_{crit}$.

Part (e): phase portrait before and after the bifurcation

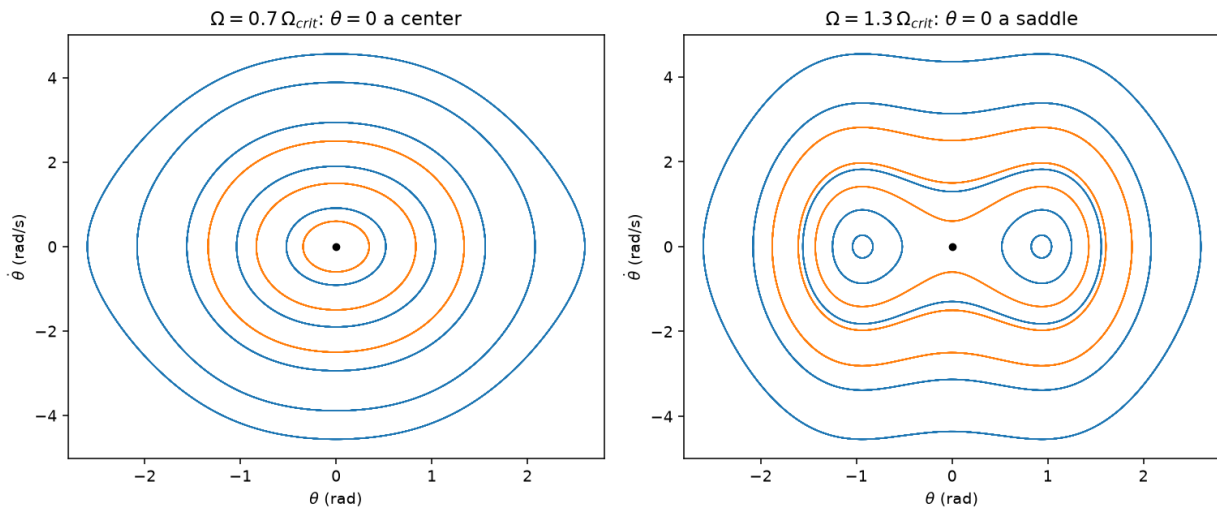
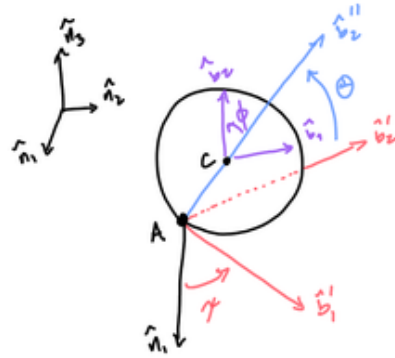


Figure 1: Phase portraits at $\Omega = 0.7 \Omega_{crit}$ and $1.3 \Omega_{crit}$ ($\ell = 1, g = 9.81$). Below Ω_{crit} : single center at $\theta = 0$. Above: pitchfork into a saddle and two centers at $\theta = \pm \arccos(\Omega_{crit}^2/\Omega^2)$.

Problem 2

Problem 2



Consider a coin rolling along its edge on a flat surface. To describe its orientation, we consider the position of the center of the coin to be defined as,

$$\vec{r}_C = x\hat{n}_1 + y\hat{n}_2 + z\hat{n}_3.$$

To describe its orientation, we use 3-1-3 Euler angles, just as in HW6. Consider the Euler angles to be:

First, imagine the coin to be laying flat on the ground. Rotate the heading of the coin by a counterclockwise angle ψ around the $\hat{n}_3$ axis. Second, rotate the coin upward around the $\hat{b}_1$ axis by the angle θ . If $\theta = 90^\circ$, the coin is standing on its edge, pointing in the direction of $\hat{b}_1$. Finally, we consider rolling around the $\hat{b}_3 = \hat{b}_3^*$ axis by the angle ϕ .

NOTE: If you're having a hard time understanding the orientation of the coin, feel free to reach out! I'll try to re-describe its orientation.

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Note: If you're having a hard time understanding the orientation of the coin, feel free to reach out! I'll try to re-describe this orientation.

- (a) Briefly, re-derive both the cosine matrix and angular velocity of the coin using the angles given (you can bring this in from HW6, and the angle definitions should match this assignment).

3-1-3 rotation matrices:

$$R(\psi) = \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad R(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix}, \quad R(\phi) = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C = R(\phi)R(\theta)R(\psi):$$

$$R(\theta)R(\psi) = \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi \cos \theta & \cos \psi \cos \theta & \sin \theta \\ \sin \psi \sin \theta & -\cos \psi \sin \theta & \cos \theta \end{bmatrix}$$

$$C = \begin{bmatrix} \cos \psi \cos \phi - \sin \psi \cos \theta \sin \phi & \sin \psi \cos \phi + \cos \psi \cos \theta \sin \phi & \sin \theta \sin \phi \\ -\cos \psi \sin \phi - \sin \psi \cos \theta \cos \phi & -\sin \psi \sin \phi + \cos \psi \cos \theta \cos \phi & \sin \theta \cos \phi \\ \sin \psi \sin \theta & -\cos \psi \sin \theta & \cos \theta \end{bmatrix}$$

$$\vec{\omega} = \dot{\psi} \hat{n}_3 + \dot{\theta} \hat{b}'_1 + \dot{\phi} \hat{b}_3$$

Note $\hat{n}_3 = \hat{b}'_3$ (the first rotation is about $\hat{n}_3$, so it is unchanged by that rotation), not $\hat{b}''_3$. Expanding $\hat{b}'_3$ in the $\{\hat{b}''\}$ basis using $R(\theta)$:

$$\hat{n}_3 = \hat{b}'_3 = \sin \theta \hat{b}''_2 + \cos \theta \hat{b}''_3$$

With $\hat{b}''_3 = \hat{b}_3$, $\hat{b}''_2 = \sin \phi \hat{b}_1 + \cos \phi \hat{b}_2$, $\hat{b}'_1 = \cos \phi \hat{b}_1 - \sin \phi \hat{b}_2$:

$$\vec{\omega} = (\dot{\theta} \cos \phi + \dot{\psi} \sin \theta \sin \phi) \hat{b}_1 + (-\dot{\theta} \sin \phi + \dot{\psi} \sin \theta \cos \phi) \hat{b}_2 + (\dot{\phi} + \dot{\psi} \cos \theta) \hat{b}_3$$

- (b) The coordinate z can be defined geometrically from one or more Euler angles. What is that relationship?

$$\boxed{z = r \sin \theta} \implies \dot{z} = r \cos \theta \dot{\theta}$$

- (c) Choosing $(x, y, \psi, \theta, \phi)$ as your five degrees of freedom, derive the Lagrangian function, $\mathcal{L}$.

$$I_1 = I_2 \equiv I_t = \frac{1}{4}Mr^2, I_3 \equiv I_a = \frac{1}{2}Mr^2:$$

$$\omega_1^2 + \omega_2^2 = \dot{\theta}^2 + \dot{\psi}^2 \sin^2 \theta, \quad \omega_3^2 = \dot{\phi}^2 + 2\dot{\phi}\dot{\psi} \cos \theta + \dot{\psi}^2 \cos^2 \theta$$

$$T = \underbrace{\frac{1}{2}M(\dot{x}^2 + \dot{y}^2 + r^2 \cos^2 \theta \dot{\theta}^2)}_{T_{trans}} + \underbrace{\frac{1}{8}Mr^2(\dot{\theta}^2 + \dot{\psi}^2 \sin^2 \theta) + \frac{1}{4}Mr^2(\dot{\phi}^2 + 2\dot{\phi}\dot{\psi} \cos \theta + \dot{\psi}^2 \cos^2 \theta)}_{T_{rot}}$$

$$\boxed{V = Mgr \sin \theta}$$

$$\mathcal{L} = T - V = \frac{1}{2}M(\dot{x}^2 + \dot{y}^2 + r^2 \cos^2 \theta \dot{\theta}^2) + \frac{1}{8}Mr^2(\dot{\theta}^2 + \dot{\psi}^2 \sin^2 \theta) + \frac{1}{4}Mr^2(\dot{\phi}^2 + 2\dot{\phi}\dot{\psi} \cos \theta + \dot{\psi}^2 \cos^2 \theta) - Mgr \sin \theta$$

- (d) There are two non-holonomic constraints on the velocity of the point that touches the ground, both the no-slip condition and a knife-edge constraint. The third constraint in the z -direction (that it stays on the ground) was taken care of in part (b). These can be taken together as the constraint, $\vec{v}_A = \vec{0}$.

Let $\vec{v}_A = \vec{v}_C + \vec{\omega} \times \vec{r}_{A/C}$, where $\vec{r}_{A/C}$ is the position of A with respect to C . Derive the constraints in the x and y direction.

$\vec{r}_{A/C} = -r\hat{b}_2''$ (fixed in $\{\hat{b}''\}$, before spin), with $C'' = R(\theta)R(\psi)$ and

$$\vec{\omega} = \dot{\theta}\hat{b}_1'' + \dot{\psi}\sin\theta\hat{b}_2'' + (\dot{\phi} + \dot{\psi}\cos\theta)\hat{b}_3''$$

$$\vec{\omega} \times \vec{r}_{A/C} = r(\dot{\phi} + \dot{\psi}\cos\theta)\hat{b}_1'' - r\dot{\theta}\hat{b}_3''$$

${}^N\vec{V}_C = -\vec{\omega} \times \vec{r}_{A/C}$, rotated into $\{\hat{n}\}$ by $(C'')^T$:

$$\dot{x} = -r(\dot{\phi} + \dot{\psi}\cos\theta)\cos\psi + r\dot{\theta}\sin\theta\sin\psi, \quad \dot{y} = -r(\dot{\phi} + \dot{\psi}\cos\theta)\sin\psi - r\dot{\theta}\sin\theta\cos\psi$$

Pfaffian form $\phi_j = \sum_i a_{ji}\dot{q}_i = 0$, $q = (x, y, \theta, \psi, \phi)$:

$$a_{1x} = 1, \quad a_{1y} = 0, \quad a_{1\theta} = -r\sin\psi\sin\theta, \quad a_{1\psi} = r\cos\psi\cos\theta, \quad a_{1\phi} = r\cos\psi$$

$$a_{2x} = 0, \quad a_{2y} = 1, \quad a_{2\theta} = r\sin\theta\cos\psi, \quad a_{2\psi} = r\sin\psi\cos\theta, \quad a_{2\phi} = r\sin\psi$$

$$\boxed{\phi_1 = \dot{x} - r\sin\theta\sin\psi\dot{\theta} + r\cos\theta\cos\psi\dot{\psi} + r\cos\psi\dot{\phi} = 0}$$

$$\boxed{\phi_2 = \dot{y} + r\sin\theta\cos\psi\dot{\theta} + r\cos\theta\sin\psi\dot{\psi} + r\sin\psi\dot{\phi} = 0}$$

- (e) Construct the matrix equation $A\ddot{y} = b$, where $y = [\dot{x}, \dot{y}, \dot{\psi}, \dot{\theta}, \dot{\phi}, \ddot{x}, \ddot{y}, \ddot{\psi}, \ddot{\theta}, \ddot{\phi}, \lambda_1, \lambda_2]^T$. The matrix A and vector b will depend on any of the state variables in the system.

As posed, y is the assembled first-order form: with $\dot{q} = (\dot{x}, \dot{y}, \dot{\psi}, \dot{\theta}, \dot{\phi})$, $\ddot{q} = (\ddot{x}, \ddot{y}, \ddot{\psi}, \ddot{\theta}, \ddot{\phi})$, and $\lambda = (\lambda_1, \lambda_2)$, the 12×12 system is the block equation

$$\begin{bmatrix} I_5 & 0_{5 \times 7} \\ 0_{7 \times 5} & A_7 \end{bmatrix} \begin{bmatrix} \dot{q} \\ \ddot{q} \\ \lambda \end{bmatrix} = \begin{bmatrix} \dot{q} \\ b_7 \end{bmatrix},$$

where the top 5 rows are just an identity block that carries the already-known $\dot{q}$ straight through, so that one solve returns the entire state derivative $(\dot{q}, \ddot{q})$ together with λ ; the bottom 7 rows are the genuine dynamics, the 7×7 system $A_7\ddot{q}_\lambda = b_7$ (with $\ddot{q}_\lambda = (\ddot{q}, \lambda)$) derived below. Because the top block is trivial, it's the 7×7 block A_7, b_7 that is actually assembled and solved each time step (`build_system`, Appendix 2).

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = \lambda_1 a_{1i} + \lambda_2 a_{2i}$$

$$\begin{aligned} x : \quad & M\ddot{x} = \lambda_1 \\ y : \quad & M\ddot{y} = \lambda_2 \\ \psi : \quad & Mr^2 \left[\frac{1}{4} (\cos^2\theta + 1) \ddot{\psi} + \frac{1}{2} \cos\theta \ddot{\phi} - \frac{1}{2} \sin\theta \dot{\theta} (\dot{\phi} + \dot{\psi}\cos\theta) \right] = r\cos\theta (\lambda_1 \cos\psi + \lambda_2 \sin\psi) \\ \theta : \quad & Mr^2 \left(\cos^2\theta + \frac{1}{4} \right) \ddot{\theta} + Mr^2 \sin\theta \cos\theta \left(\frac{1}{4} \dot{\psi}^2 - \dot{\theta}^2 \right) + \frac{1}{2} Mr^2 \sin\theta \dot{\phi} \dot{\psi} + Mgr \cos\theta \\ & = r\sin\theta (\lambda_2 \cos\psi - \lambda_1 \sin\psi) \\ \phi : \quad & \frac{1}{2} Mr^2 (\ddot{\phi} + \ddot{\psi}\cos\theta - \dot{\psi}\dot{\theta}\sin\theta) = r (\lambda_1 \cos\psi + \lambda_2 \sin\psi) \end{aligned}$$

Differentiating ϕ_1, ϕ_2 once:

$$\boxed{\ddot{x} - r\sin\theta\sin\psi\ddot{\theta} + r\cos\theta\cos\psi\ddot{\psi} + r\cos\psi\ddot{\phi} = r\cos\theta\sin\psi(\dot{\theta}^2 + \dot{\psi}^2) + 2r\sin\theta\cos\psi\dot{\theta}\dot{\psi} + r\sin\psi\dot{\phi}\dot{\psi}}$$

$$\ddot{y} + r \sin \theta \cos \psi \ddot{\theta} + r \cos \theta \sin \psi \ddot{\psi} + r \sin \psi \ddot{\phi} = -r \cos \theta \cos \psi (\dot{\theta}^2 + \dot{\psi}^2) + 2r \sin \theta \sin \psi \dot{\theta} \dot{\psi} - r \cos \psi \dot{\phi} \dot{\psi}$$

This gives 7 equations in the 7 unknowns $\ddot{q}_\lambda = (\ddot{x}, \ddot{y}, \ddot{\psi}, \ddot{\theta}, \ddot{\phi}, \lambda_1, \lambda_2)$. Writing them as $A_7 \ddot{q}_\lambda = b_7$ with that ordering, matching the professor's (ψ, θ) order rather than (θ, ψ) (shorthand $c_\theta = \cos \theta$, $s_\theta = \sin \theta$, $c_\psi = \cos \psi$, $s_\psi = \sin \psi$; all unlabeled entries are 0):

$$A_7 = \begin{bmatrix} M & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & M & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & \frac{1}{4}Mr^2(c_\theta^2 + 1) & 0 & \frac{1}{2}Mr^2c_\theta & -rc_\theta c_\psi & -rc_\theta s_\psi \\ 0 & 0 & 0 & Mr^2(c_\theta^2 + \frac{1}{4}) & 0 & rs_\theta s_\psi & -rs_\theta c_\psi \\ 0 & 0 & \frac{1}{2}Mr^2c_\theta & 0 & \frac{1}{2}Mr^2 & -rc_\psi & -rs_\psi \\ 1 & 0 & rc_\psi c_\theta & -rs_\psi s_\theta & rc_\psi & 0 & 0 \\ 0 & 1 & rs_\psi c_\theta & rs_\theta c_\psi & rs_\psi & 0 & 0 \end{bmatrix}$$

$$b_7 = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{2}Mr^2s_\theta\dot{\theta}(\dot{\phi} + \dot{\psi}c_\theta) \\ -Mr^2s_\theta c_\theta(\frac{1}{4}\dot{\psi}^2 - \dot{\theta}^2) - \frac{1}{2}Mr^2s_\theta\dot{\phi}\dot{\psi} - Mgrc_\theta \\ \frac{1}{2}Mr^2\dot{\psi}\dot{\theta}s_\theta \\ rs_\psi c_\theta(\dot{\psi}^2 + \dot{\theta}^2) + rs_\psi\dot{\phi}\dot{\psi} + 2rs_\theta c_\psi\dot{\psi}\dot{\theta} \\ 2rs_\psi s_\theta\dot{\psi}\dot{\theta} - rc_\psi c_\theta(\dot{\psi}^2 + \dot{\theta}^2) - rc_\psi\dot{\phi}\dot{\psi} \end{bmatrix}$$

with rows/columns ordered $(\ddot{x}, \ddot{y}, \ddot{\psi}, \ddot{\theta}, \ddot{\phi}, \lambda_1, \lambda_2)$; rows 1–2 are the x, y equations, rows 3–5 are the ψ, θ, ϕ equations above, and rows 6–7 are the differentiated constraints $\dot{\phi}_1 = 0$, $\dot{\phi}_2 = 0$.

- (f) Using this matrix equation to numerically solve your equations of motion. Choose a reasonable mass and radius, and choose an initial condition with $\theta_0 \neq 0$ and $\dot{\psi}_0 \neq 0$, and choose $\dot{x}_0, \dot{y}_0$ based on your equations of constraint. Otherwise choose initial conditions equal to 0.

$r = 1$, $M = 1$; $\theta_0 = 1.2$, $\psi_0 = \phi_0 = 0$, $\dot{\theta}_0 = 0$, $\dot{\psi}_0 = 1.5$, with $\dot{x}_0, \dot{y}_0$ from part (d).

The problem says otherwise to take initial conditions equal to 0, so I first ran that literal case, $\dot{\phi}_0 = 0$, to check it. With no spin the coin still nutates periodically (Figure 2, period ≈ 1.5 s), but it's large-amplitude: θ repeatedly swings all the way from 1.2 down to 0.151 rad, i.e. $z = r \sin \theta$ repeatedly drops from 0.932 to 0.151. The coin is being carried down close to lying flat on every cycle, the point at which the point-contact rolling-on-edge model stops describing the physical situation well. So $\dot{\phi}_0 = 0$ is not a meaningful rolling initial condition here, and a nonzero spin is needed to keep the nutation to a small, edge-like range.

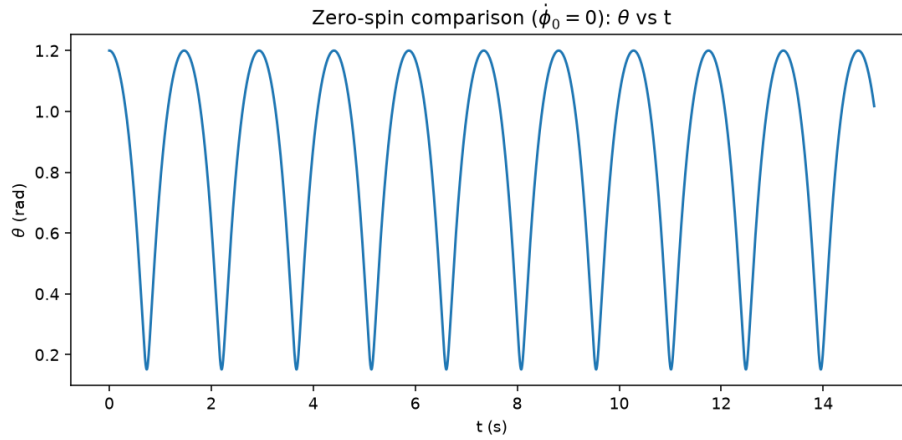


Figure 2: Zero-spin comparison ($\dot{\phi}_0 = 0$): θ nutates periodically (period ≈ 1.5 s) between 0.151 and 1.2 rad.

Taking instead $\dot{\phi}_0 = 6$, the coin nutates about the initial tilt rather than falling toward flat. Running the integration prints

$$\text{max relative drift in energy } E : 1.418 \times 10^{-12}, \quad \text{max } |\phi_1|, \text{max } |\phi_2| : 1.104 \times 10^{-11}, 1.285 \times 10^{-11},$$

confirming energy conservation and that the rolling constraints $\phi_1 = \phi_2 = 0$ are satisfied to numerical precision throughout the integration.

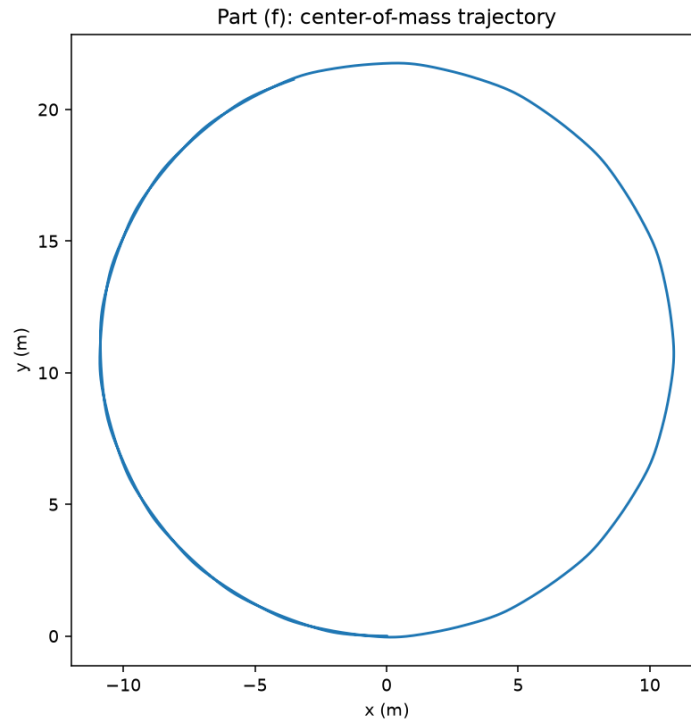


Figure 3: Part (f): center-of-mass trajectory ($\dot{\phi}_0 = 6$).

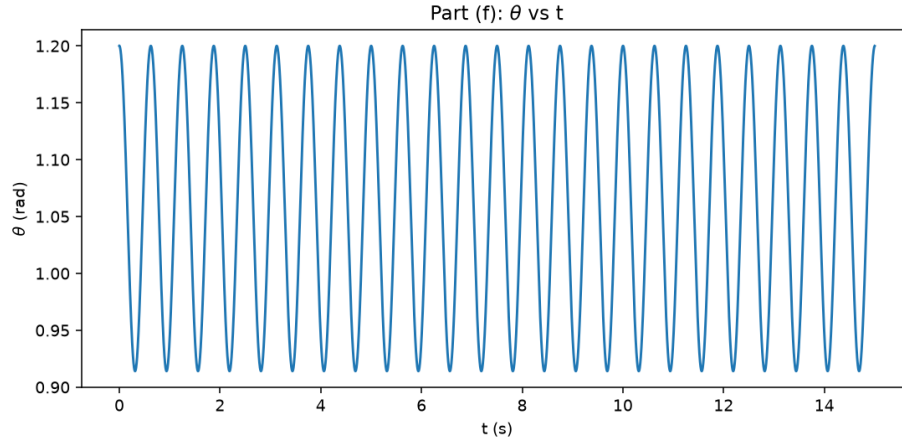


Figure 4: Part (f): θ vs. t ($\dot{\phi}_0 = 6$).

Nutates between $\theta \approx 0.92$ and 1.2 rad while rolling on a large circular arc of radius ≈ 10.9 m (fit to the trajectory above).

That radius is worth checking against $v = r(\dot{\phi} + \dot{\psi} \cos \theta)$. Doing this with the instantaneous $v(0) \approx 6.5$ m/s and the initial rate $\dot{\psi}_0 = 1.5$ rad/s is not a valid comparison: $\dot{\psi}$ oscillates with θ rather than staying constant, so mixing one instant of v with a different, time-averaged rate isn't a like-for-like check (it happens to land near 4.4 m, off from the fitted radius by more than a factor of two). The consistent version compares two time-averaged quantities: the total arc length traveled, $\int v dt \approx 99.0$ m, against the total heading change over the same run, $|\Delta\psi| \approx 9.10$ rad, giving $\int v dt / |\Delta\psi| \approx 10.9$ m, matching the fitted radius of 10.9 m.

Note that the transverse moment of inertia for a disk is $I_t = \frac{1}{4}Mr^2$, and the axial moment of inertia is $I_a = \frac{1}{2}Mr^2$.

Appendix: Code Listings

Problem 1 Code

```

1 import os
2 import matplotlib
3 matplotlib.use('Agg')
4 import numpy as np
5 from scipy.integrate import solve_ivp
6 import matplotlib.pyplot as plt
7
8 FIG_DIR = os.path.join(os.path.dirname(__file__), '..', 'latex', 'Figures', 'Graphs', 'Problem1')
9 os.makedirs(FIG_DIR, exist_ok=True)
10
11 g, l = 9.81, 1.0
12 Omega_crit = np.sqrt(48/79 * g/l)
13 print(f"Omega_crit = {Omega_crit:.4f} rad/s")
14
15
16 def theta_ddot(theta, Omega):
17     return (79/81)*Omega**2*np.sin(theta)*np.cos(theta) - (16/27)*(g/l)*np.sin(theta)
18
19
20 def deriv(t, state, Omega):
21     theta, thetadot = state
22     return [thetadot, theta_ddot(theta, Omega)]
23
24
25 def phase_portrait(ax, Omega, title):
26     t_span = [0, 20]
27     t_eval = np.linspace(*t_span, 4000)
28
29     # librating orbits released from rest at various theta0
30     for theta0 in np.linspace(-2.6, 2.6, 11):
31         if abs(theta0) < 1e-3:
32             continue
33         sol = solve_ivp(deriv, t_span, [theta0, 0.0], args=(Omega,),
34                         t_eval=t_eval, method='DOP853', rtol=1e-10, atol=1e-12)
35         ax.plot(sol.y[0], sol.y[1], color='C0', lw=0.8)
36
37     # trajectories launched from theta=0 with a range of angular velocities,
38     # tracing out the separatrix and the circulating orbits beyond it
39     for thetadot0 in [-2.5, -1.5, -0.6, 0.6, 1.5, 2.5]:
40         sol = solve_ivp(deriv, t_span, [0.0, thetadot0], args=(Omega,),
41                         t_eval=t_eval, method='DOP853', rtol=1e-10, atol=1e-12)
42         ax.plot(sol.y[0], sol.y[1], color='C1', lw=0.8)
43
44     ax.plot(0, 0, 'ko', ms=4)
45     ax.set_xlabel(r'$\theta$ (rad)')
46     ax.set_ylabel(r'$\dot{\theta}$ (rad/s)')
47     ax.set_title(title)
48     ax.set_xlim(-2.8, 2.8)
49
50
51 fig, axes = plt.subplots(1, 2, figsize=(11, 5))
52 phase_portrait(axes[0], 0.7*Omega_crit, r'$\Omega=0.7\backslash\Omega_{crit}$: $\theta=0$ a center')
53 phase_portrait(axes[1], 1.3*Omega_crit, r'$\Omega=1.3\backslash\Omega_{crit}$: $\theta=0$ a saddle')
54 fig.suptitle('Part (e): phase portrait before and after the bifurcation')

```

```

55 fig.tight_layout()
56 fig.savefig(os.path.join(FIG_DIR, 'partE_phase_portrait.png'), dpi=140)
57 print("done")

```

Listing 1: Part (e) phase-portrait integration, sweeping initial conditions above and below Ω_{crit} .

Problem 2 Code

```

1  import os
2  import matplotlib
3  matplotlib.use('Agg')
4  import numpy as np
5  from scipy.integrate import solve_ivp
6  from scipy.signal import find_peaks
7  import matplotlib.pyplot as plt
8
9  FIG_DIR = os.path.join(os.path.dirname(__file__), '..', 'latex', 'Figures', 'Graphs', 'Problem2')
10 os.makedirs(FIG_DIR, exist_ok=True)
11
12 g = 9.81
13
14
15 def build_system(state, r, M):
16     """Assembles the 7x7 dynamics block A_7 (xdd, ydd, psdd, thdd, phdd, lam1, lam2 = np.linalg.
17         solve(A_7, b_7)).
18     This is the only nontrivial block of the full 12x12 system in part (e); the other 5 rows
19     (qdot -> qdot) are just an identity pass-through and aren't worth assembling as a linear solve.
20     """
21
22     th, ps, ph, thd, psd, phd = state
23     c, s = np.cos, np.sin
24     ct, st = c(th), s(th)
25     cp, sp = c(ps), s(ps)
26
27     A = np.zeros((7, 7))
28     b = np.zeros(7)
29
30     # x: M*xdd = lam1
31     A[0, 0] = M
32     A[0, 5] = -1.0
33
34     # y: M*ydd = lam2
35     A[1, 1] = M
36     A[1, 6] = -1.0
37
38     # psi
39     A[2, 2] = M * r**2 * (ct**2 + 1) / 4
40     A[2, 4] = M * r**2 * ct / 2
41     A[2, 5] = -r * ct * cp
42     A[2, 6] = -r * ct * sp
43     b[2] = M * r**2 * (st * thd / 2) * (phd + psd * ct)
44
45     # theta
46     A[3, 3] = M * r**2 * (ct**2 + 0.25)
47     A[3, 5] = r * st * sp
48     A[3, 6] = -r * st * cp
49     b[3] = (-M * r**2 * st * ct * (psd**2 / 4 - thd**2)

```

```

48         - 0.5 * M * r**2 * st * phd * psd
49         - M * g * r * ct)
50
51     # phi
52     A[4, 2] = M * r**2 * ct / 2
53     A[4, 4] = M * r**2 / 2
54     A[4, 5] = -r * cp
55     A[4, 6] = -r * sp
56     b[4] = M * r**2 / 2 * psd * thd * st
57
58     # d/dt(phi1) = 0
59     A[5, 0] = 1.0
60     A[5, 2] = r * cp * ct
61     A[5, 3] = -r * sp * st
62     A[5, 4] = r * cp
63     b[5] = (r * sp * ct * (psd**2 + thd**2) + r * sp * phd * psd
64             + 2 * r * st * cp * psd * thd)
65
66     # d/dt(phi2) = 0
67     A[6, 1] = 1.0
68     A[6, 2] = r * sp * ct
69     A[6, 3] = r * st * cp
70     A[6, 4] = r * sp
71     b[6] = (2 * r * sp * st * psd * thd - r * cp * ct * (psd**2 + thd**2)
72             - r * cp * phd * psd)
73
74     return A, b
75
76
77 def deriv(t, state, r, M):
78     x, y, th, ps, ph, xd, yd, thd, psd, phd = state
79     A, b = build_system((th, ps, ph, thd, psd, phd), r, M)
80     xdd, ydd, psdd, thdd, phdd, lam1, lam2 = np.linalg.solve(A, b)
81     return [xd, yd, thd, psd, phd, xdd, ydd, thdd, psdd, phdd]
82
83
84 r, M = 1.0, 1.0
85
86 theta0, psi0, phi0 = 1.2, 0.0, 0.0
87 thetadot0, psidot0 = 0.0, 1.5
88 x0, y0 = 0.0, 0.0
89
90
91 def constrained_ic(phidot0):
92     xdot0 = -r * (phidot0 + psidot0 * np.cos(theta0)) * np.cos(psi0) + r * thetadot0 * np.sin(
93         theta0) * np.sin(psi0)
94     ydot0 = -r * (phidot0 + psidot0 * np.cos(theta0)) * np.sin(psi0) - r * thetadot0 * np.sin(
95         theta0) * np.cos(psi0)
96     return [x0, y0, theta0, psi0, phi0, xdot0, ydot0, thetadot0, psidot0, phidot0]
97
98 t_end = 15.0
99 t_eval = np.linspace(0, t_end, 6000)
100
101 # --- Zero-spin comparison: same theta0, psidot0, but phidot0 = 0 ---
102 state0_zerospin = constrained_ic(0.0)
103 sol0 = solve_ivp(deriv, [0, t_end], state0_zerospin, args=(r, M), t_eval=t_eval,
104                 method='DOP853', rtol=1e-11, atol=1e-13)
105 theta_zerospin = sol0.y[2]

```

```

105 troughs, _ = find_peaks(-theta_zerospin)
106 period_zerospin = np.mean(np.diff(sol0.t[troughs]))
107 print(f"phidot0=0: theta oscillates periodically between {theta_zerospin.min():.3f} and {
      theta_zerospin.max():.3f} rad, "
108       f"period approx {period_zerospin:.2f} s "
109       f"(z=r*sin(theta) between {r*np.sin(theta_zerospin.min()):.3f} and {r*np.sin(theta_zerospin.
      max():.3f})")
110
111 fig, ax = plt.subplots(figsize=(8, 4))
112 ax.plot(sol0.t, theta_zerospin)
113 ax.set_xlabel('t (s)'); ax.set_ylabel(r'$\theta$ (rad)')
114 ax.set_title(r'Zero-spin comparison ($\dot{\phi}_0=0$): $\theta$ vs t')
115 fig.tight_layout(); fig.savefig(os.path.join(FIG_DIR, 'partF_zerospin_theta_vs_t.png'), dpi=140)
116
117 # --- Main run: phidot0 = 6, giving a meaningful rolling/nutating solution ---
118 phidot0 = 6.0
119 state0 = constrained_ic(phidot0)
120
121 sol = solve_ivp(deriv, [0, t_end], state0, args=(r, M), t_eval=t_eval,
122               method='DOP853', rtol=1e-11, atol=1e-13, dense_output=True)
123 x, y, theta, psi, phi, xd, yd, thd, psd, phd = sol.y
124
125
126 It, Ia = 0.25 * M * r**2, 0.5 * M * r**2
127 ct, st = np.cos(theta), np.sin(theta)
128 cph, sph = np.cos(phi), np.sin(phi)
129 cps, sps = np.cos(psi), np.sin(psi)
130 w1 = thd * cph + psd * st * sph
131 w2 = -thd * sph + psd * st * cph
132 w3 = phd + psd * ct
133 T = 0.5 * M * (xd**2 + yd**2 + (r * ct * thd)**2) + 0.5 * It * (w1**2 + w2**2) + 0.5 * Ia * w3**2
134 V = M * g * r * st
135 E = T + V
136 print(f"max relative drift in energy E: {np.max(np.abs(E - E[0])) / abs(E[0]):.3e}")
137
138 phi1 = xd + r * (phd + psd * ct) * cps - r * thd * st * sps
139 phi2 = yd + r * (phd + psd * ct) * sps + r * thd * st * cps
140 print(f"max |phi1|, |phi2| (rolling constraint residual): {np.max(np.abs(phi1)):.3e}, {np.max(np.
      abs(phi2)):.3e}")
141
142 v0 = r * abs(phidot0 + psidot0 * np.cos(theta0))
143 speed = np.sqrt(xd**2 + yd**2)
144 print(f"v at t=0: {v0:.3f} m/s; mean speed over run: {speed.mean():.3f} m/s")
145
146 # arc length / total heading change: both time-averaged, so the instantaneous v0
147 # is never mixed with a time-averaged rate
148 trapz = np.trapezoid if hasattr(np, 'trapezoid') else np.trapz
149 arc_length = trapz(speed, sol.t)
150 delta_psi = psi[-1] - psi[0]
151 R_arc = arc_length / abs(delta_psi)
152 print(f"arc length: {arc_length:.3f} m; total heading change |Delta psi|: {abs(delta_psi):.3f} rad;
      "
153       f"R = arc_length/|Delta psi| = {R_arc:.3f} m")
154 print(f"mean psidot over run: {psd.mean():.3f} rad/s; mean(v)/mean(psidot) = {speed.mean()/psd.mean
      ():.3f} m")
155
156 # algebraic circle fit (Kasa) to the x,y trajectory, for the actual turning radius
157 Amat = np.column_stack([x, y, np.ones_like(x)])
158 bvec = x**2 + y**2

```

```

159 sol_fit, *_ = np.linalg.lstsq(Amat, bvec, rcond=None)
160 xc, yc = sol_fit[0] / 2, sol_fit[1] / 2
161 R_fit = np.sqrt(sol_fit[2] + xc**2 + yc**2)
162 print(f"circle fit: center=({xc:.3f}, {yc:.3f}), radius={R_fit:.3f} m")
163
164 fig, ax = plt.subplots(figsize=(6, 6))
165 ax.plot(x, y)
166 ax.set_xlabel('x (m)'); ax.set_ylabel('y (m)'); ax.set_aspect('equal')
167 ax.set_title('Part (f): center-of-mass trajectory')
168 fig.tight_layout(); fig.savefig(os.path.join(FIG_DIR, 'partF_xy_trajectory.png'), dpi=140)
169
170 fig, ax = plt.subplots(figsize=(8, 4))
171 ax.plot(sol.t, theta)
172 ax.set_xlabel('t (s)'); ax.set_ylabel(r'$\theta$ (rad)')
173 ax.set_title(r'Part (f): $\theta$ vs t')
174 fig.tight_layout(); fig.savefig(os.path.join(FIG_DIR, 'partF_theta_vs_t.png'), dpi=140)
175 print("done")

```

Listing 2: Integrates the coin's equations of motion via the part (e) Lagrange-multiplier system, with energy and constraint checks.